DL TUMOR CLASSIFICATION

Comprehensive Analysis, External Validation, and Project Transfer Record

Brain Tumor MRI Classification | Status snapshot: 3 October 2026

MRI 224x224	CNN backbone	Spatial grid -> sequence	LSTM	BN + Dropout	Dense128 + ReLU	Neural head

Deep features -> StandardScaler (fit train only) -> PCA64 (fit train only) -> Linear SVM

5	40	70/70	15	2,644	1,648
backbones	fixed 30-epoch runs	Phase 2B hybrid evals	frozen pipelines	full external cohort	balanced subset

Scientific interpretation gate
The original internal train/validation/test arrangement contains material cross-split duplication, so historical internal results remain contamination-aware rather than clean independence claims. External validation is now complete on a BDNeuro V8 cohort after excluding 3,297 exact decoded-pixel overlaps with the original 7,023-image dataset. Residual perceptual similarity was screened but not used as an automatic exclusion rule. Patient-level independence remains unknown.

Purpose: a self-contained reference for future ChatGPT/Claude work, report writing, Review-II preparation, methodology continuity, and viva discussion.

How to use this document

This report updates and supersedes the 2 October 2026 analysis record. It preserves the established Phase 1, leakage, GUI, explainability and Phase 2 training history, then adds the completed downstream PCA+SVM evaluations, the 15-pipeline pre-external freeze, the BDNeuro external-dataset audit, exact-overlap cleaning, balanced sensitivity subset, and the completed external-validation results.

What changed since the 2 October report

Item	2 Oct status	3 Oct status
Phase 2A downstream hybrid evaluation	Partially consolidated	10/10 retained-state evaluations complete
Phase 2B downstream hybrid evaluation	Training complete; full hybrid pending	70/70 evaluations complete; 0 failed
Pre-external model freeze	Planned	15 exact pipelines frozen and artifact-hashed
External dataset	Not selected	BDNeuro V8 selected, audited and exact-overlap cleaned
Primary external cohort	Pending	2,644 images
Balanced sensitivity cohort	Pending	1,648 images, 412/class
External 15-model evaluation	Pending	Complete for both full and balanced cohorts
Web-image stress test	Not defined	Protocol defined: 40 source-verified images; still pending

Evidence hierarchy used in this update

Tier	Evidence	Use
A	Current 15-model external master CSV outputs supplied 3 Oct 2026	External accuracy, macro metrics and internal-to-external gaps
B	Current terminal/audit outputs and generated cohort manifests	BDNeuro counts, exact-overlap removal, balanced subset, freeze status
C	Earlier 2 Oct comprehensive report and established project artifacts	Phase 1, leakage, GUI, Grad-CAM, occlusion, Phase 2 training history

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1. Executive summary

This project implements and extends a hybrid four-class brain-tumor MRI classification pipeline using five pretrained CNN backbones connected to an LSTM and a downstream StandardScaler + PCA64 + linear SVM classifier. The classes are glioma, meningioma, no tumor and pituitary. The locked backbones are VGG16, VGG19, ResNet50, ResNet152 and EfficientNet-B3.

Current stage
Phase 1, leakage audit, material-clean evaluation, GUI, Grad-CAM, final-SVM occlusion, Phase 2A training, Phase 2B training, all retained-state downstream PCA+SVM evaluations, 15-pipeline pre-external freeze, BDNeuro V8 overlap audit, exact-clean cohort construction, deterministic balanced subset creation, and two external 15-model evaluations are COMPLETE. The 40-image web stress test and duplicate-group-aware internal re-splitting/retraining remain future work.

Most important new findings

- Phase 2B downstream evaluation is fully complete: 35 configurations x 2 retained states = 70 full CNN-LSTM -> scaler -> PCA -> SVM evaluations; 68 were newly completed, 2 already existed, and 0 failed.

- The 15 external pipelines were frozen before external evaluation: one Phase 1, one Phase 2A, and one Phase 2B representative for each of five backbones. Selection used internal full-hybrid validation only; external results were not used for reselection.

- BDNeuro V8 contained 5,941 class-folder MRI images, but direct cross-dataset auditing found 3,297 unique exact decoded-pixel overlaps with the original 7,023-image development corpus. Those exact overlaps were excluded.

- The primary external cohort contains 2,644 exact-pixel-clean images: 1,159 glioma, 472 meningioma, 412 notumor and 601 pituitary. A deterministic balanced sensitivity subset contains 1,648 images, exactly 412 per class.

- Across all 15 frozen pipelines, full external accuracy ranged from 75.00% to 88.31%; balanced-subset accuracy ranged from 77.00% to 90.47%. Every model showed an internal-to-external accuracy drop, demonstrating a substantial domain-generalization gap.

- ResNet152 Phase 2A produced the highest measured external accuracy in both cohorts: 88.31% full-cohort accuracy / 87.85% macro F1 and 90.47% balanced accuracy / 90.50% macro F1.

- Phase 2B tuning did not consistently transfer externally. It improved ResNet50 strongly and EfficientNet-B3 modestly on the balanced subset, but reduced external accuracy relative to Phase 2A for VGG16, VGG19 and ResNet152.

- On average across five backbones, Phase 2A generalized better than Phase 2B: 83.21% vs 82.34% full-cohort accuracy and 85.39% vs 84.33% balanced accuracy.

Current workstream status

Workstream	Status	Meaning
Phase 1 five-backbone baseline	COMPLETE	Neural head + full PCA/SVM pipeline
Internal duplicate/leakage audit	COMPLETE	562 material cross-split pairs; patient independence unknown
Material-clean internal test evaluation	COMPLETE	n=1,061; post-hoc only
Phase 2A fixed-30 training	COMPLETE	5 matched baselines
Phase 2A downstream retained-state evaluation	COMPLETE	10/10 states
Phase 2B DL sensitivity training	COMPLETE	35/35 variants
Phase 2B downstream retained-state evaluation	COMPLETE	70/70 states
15-pipeline freeze	COMPLETE	Hashed manifest written before external evaluation
BDNeuro external audit + clean cohort	COMPLETE	5,941 -> 2,644 after exact-pixel overlap exclusion
Full external validation	COMPLETE	15 frozen pipelines x 2,644 images
Balanced sensitivity analysis	COMPLETE	15 frozen pipelines x 1,648 images
Web-image stress test	PENDING	40 source-verified images, 10/class
Duplicate-group-aware internal retraining	UNRESOLVED	Stronger remediation if clean internal-generalization claim is required

2. Locked scope, architecture and data

2.1 Classification and model scope

- **Input:** ordinary MRI image files; DICOM remains outside current scope.

- **Classes:** glioma, meningioma, notumor, pituitary.

- **Backbones:** VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3.

- **Core neural path:** MRI -> CNN spatial feature map -> spatial sequence -> LSTM -> BatchNorm -> Dropout -> Dense128/ReLU -> temporary neural classifier.

- **Final standardized classifier:** 128-D deep features -> StandardScaler -> PCA(64) -> linear SVM.

- **Training baseline:** RMSprop, LR 1e-4, batch 8, LSTM 256, dropout 0.3, raw-logit CrossEntropyLoss, train-only augmentation, deterministic validation/test/feature extraction.

- **Transfer-learning asymmetry:** VGG16/VGG19 convolutional backbones are frozen for stability after unfrozen VGG collapse; ResNet/EfficientNet belong to the fine-tuned family.

2.2 Original dataset and saved split

Partition	Images	Role
Original Kaggle Training folder	5,712	Source of fixed train/validation split
Saved training subset	4,569	CNN-LSTM optimization; fit scaler/PCA/SVM
Saved validation subset	1,143	Checkpoint/model selection and Phase 2 sensitivity
Official Kaggle Testing folder	1,311	Historical final evaluation
Post-hoc material_clean test	1,061	Leakage-aware subset after excluding 250 implicated test images

Interpretation rule
StandardScaler, PCA and SVM are always fit on TRAIN only. Validation, internal test and external data are transformed only. The external BDNeuro cohorts were never used for fitting, tuning or checkpoint selection.

3. Project journey and methodology evolution

Milestone	What changed / why
Base-paper implementation	Hybrid CNN+LSTM+SVM paper chosen as implementation target; faithful base first, extensions later.
Framework change	TensorFlow path abandoned after RTX 5070 / Blackwell PTX incompatibility; moved to PyTorch under Ubuntu/WSL2.
ResNet152 staged prototype	CNN feature map, spatial sequence, LSTM, post-LSTM layers, full forward/training step, then full training.
Generic five-backbone framework	Unified CNNLSTM implementation for VGG16, VGG19, ResNet50, ResNet152 and EfficientNet-B3.
VGG stabilization	Fully unfrozen VGG collapsed; pretrained VGG convolutional backbones frozen.
Standardized hybrid evaluation	Deterministic deep features -> scaler -> PCA64 -> linear SVM.
GUI + explainability	Streamlit GUI; Grad-CAM for neural head; occlusion for final SVM path.
Internal leakage audit	Material cross-split duplication discovered; interpretation of historical scores revised.
Phase 2A	Five fixed 30-epoch baseline runs to study convergence/stability.
Phase 2B	35 one-factor hyperparameter variants; then both retained states evaluated through full hybrid path.
Pre-external freeze	15 pipelines fixed and SHA256-hashed before any external evaluation.
BDNeuro audit	External candidate directly compared against all 7,023 internal images; 3,297 unique exact-pixel overlaps found.
External validation	Exact-overlap-clean full cohort and deterministic balanced subset evaluated using all 15 frozen pipelines.

4. Phase 1 baseline and leakage-aware interpretation

4.1 Neural-head vs PCA+SVM downstream-classifier comparison

The correct term is **neural-head vs PCA+SVM downstream-classifier comparison**. It is not a PCA-only ablation because dimensionality reduction and classifier family change together.

Backbone	Neural test acc.	Neural macro F1	PCA+SVM test acc.	Accuracy change
VGG16	93.67%	93.32%	94.97%	+1.30 pp
VGG19	93.14%	92.72%	93.52%	+0.38 pp
ResNet50	97.33%	97.19%	98.09%	+0.76 pp
ResNet152	98.17%	98.06%	98.40%	+0.23 pp
EfficientNet-B3	98.02%	97.94%	98.47%	+0.45 pp

4.2 Internal cross-split duplicate/leakage audit

Audit item	Result
Images audited	7,023
Exact cross-split duplicate pairs	464
High-confidence near-duplicate pairs	98
Material cross-split pairs	562
Additional perceptual review candidates	481
Unique official-test images implicated	250
material_clean test size	1,061 of 1,311
Patient-level independence	Cannot be established without reliable patient identifiers

4.3 Phase 1 material-clean test results

Backbone	material_clean accuracy	material_clean macro F1	Original test accuracy
VGG16	94.816%	94.302%	94.966%
VGG19	93.025%	92.415%	93.516%
ResNet50	98.303%	98.134%	98.093%
ResNet152	98.398%	98.315%	98.398%
EfficientNet-B3	98.680%	98.511%	98.474%

What this means
Removing implicated test images did not collapse performance, but material_clean is still a post-hoc subset. It does not repair duplicated groups used during training/validation and does not prove patient-level independence.

5. GUI and explainability

5.1 Streamlit GUI

- Select any of the five standardized backbones and upload an MRI image.

- Display final SVM prediction, neural-head comparison, agreement status, scores, runtime and inference details.

- Display validation/test metrics, confusion matrices, per-class results, training curves and cross-backbone comparison.

- Load saved scaler/PCA/SVM artifacts; nothing is refitted on an uploaded image.

- Research/educational prototype only; not a diagnostic medical device.

5.2 Grad-CAM evaluation

Formal ResNet152 Grad-CAM review used 20 fixed official-test images (5/class) and compared Layer 3, Layer 4 and a 50:50 fused map. Correct classification often coexisted with poor or off-lesion localization; hotspots frequently involved skull margin, face, border or extracranial regions. Grad-CAM therefore remains a coarse qualitative contribution map, not segmentation or proof of clinically valid reasoning.

Map	Good	Partial	Poor	Chosen most relevant
Layer 3 (14x14)	1	9	5	9/15 tumor images
Layer 4 (7x7)	3	2	10	6/15
Fused 50:50	2	6	7	0/15

5.3 Final-SVM occlusion comparison

Eleven fixed cases were evaluated using 32x32 patches, stride 16 and 169 occlusions/image through the complete CNN-LSTM -> scaler -> PCA -> SVM path. Among 9 tumor cases, localization was Good 2, Partial 2, Poor 5. Across 1,859 occlusions there were 58 class flips (3.12% pooled); median per-image flip rate was 0%. Occlusion and Grad-CAM answer different questions and neither should be interpreted as tumor segmentation.

6. Phase 2A - common fixed 30-epoch study

Phase 2A removed early termination as a confounder. Each baseline configuration was trained for exactly 30 epochs. The purpose was convergence/stability comparison rather than new final-test selection.

Backbone	LR	LSTM	Dropout	Best ep.	Best neural val	Epoch30 neural val
VGG16	1e-4	256	0.3	30	95.45%	95.45%
VGG19	1e-4	256	0.3	30	95.19%	95.19%
ResNet50	1e-4	256	0.3	30	98.43%	98.43%
ResNet152	1e-4	256	0.3	10	98.78%	98.69%
EfficientNet-B3	1e-4	256	0.3	27	99.30%	98.69%

6.1 Phase 2A downstream full-hybrid evaluation

Both retained checkpoints - best_val_within_30 and epoch_30 - were evaluated through deterministic feature extraction, training-only scaler/PCA/SVM fitting and validation scoring. All 10 retained-state evaluations are complete. For the frozen external comparison, the representative state for each backbone was chosen by internal full-hybrid validation accuracy, then macro F1; an exact tie favored best_val_within_30.

Backbone	Selected state	Epoch	Hybrid val acc.	Hybrid val macro F1
VGG16	best_val_within_30	30	95.888%	95.746%
VGG19	best_val_within_30	30	95.626%	95.462%
ResNet50	best_val_within_30	30	98.688%	98.644%
ResNet152	best_val_within_30	10	98.600%	98.545%
EfficientNet-B3	best_val_within_30	27	99.300%	99.270%

Why both retained states were evaluated
The downstream PCA+SVM representation can rank checkpoints differently from the temporary neural head. Therefore the best neural checkpoint was not assumed to be the best downstream representation.

7. Phase 2B - sensitivity and full-hybrid downstream evaluation

7.1 Training design

Phase 2B used seven one-factor perturbations around LR 1e-4 / LSTM 256 / dropout 0.3 for each of five backbones: LR 1e-5, 5e-5, 5e-4; LSTM 128, 512; dropout 0.2, 0.5. This produced 35 new fixed-30-epoch runs plus the five Phase 2A baselines = 40 matched runs.

Finding	Training-level evidence
Learning-rate sensitivity dominates	Mean within-backbone peak-validation range 4.62 pp vs 0.58 pp LSTM width and 0.42 pp dropout.
LR preference is architecture-dependent	VGGs tolerated 5e-4 relatively well; ResNets degraded strongly at 5e-4; EfficientNet-B3 baseline 1e-4 remained strongest among LR-only settings.
LSTM512 gave modest peak gains	All five backbones increased peak neural validation vs LSTM256 by +0.09 to +0.70 pp, but late-epoch stability varied.
Dropout effects smaller/inconsistent	No one dropout value was preferred by all architectures.

7.2 Complete Phase 2B downstream PCA+SVM evaluation

Each of the 35 configurations retained two checkpoints: best_val_within_30 and epoch_30. Both were evaluated through the full downstream hybrid path: 35 x 2 = 70 evaluations. Queue result: 68 newly completed, 2 already existed, 0 failed, 70/70 accounted for.

Key methodological result
Across the 35 configurations, the best-neural retained state had higher downstream SVM validation accuracy in 20 cases, epoch 30 was higher in 8, and 7 were identical. This supports evaluating both retained states. It does NOT establish the absolute best downstream epoch among all epochs 1-30 because only two states were retained.

7.3 Phase 2B representative per backbone for the frozen comparison

Backbone	Changed configuration	State	Epoch	Hybrid val acc.	Macro F1
VGG16	lr_5e-4	best	27	96.500%	96.321%
VGG19	dropout_0.5	epoch30	30	95.538%	95.382%
ResNet50	dropout_0.2	best	28	99.125%	99.122%
ResNet152	lstm_512	best	26	99.300%	99.265%
EfficientNet-B3	lstm_512	best	29	99.388%	99.377%

Important nuance: VGG19's best changed Phase 2B configuration was slightly below its Phase 2A baseline internally. It was still frozen as the Phase 2B changed-configuration representative so the external comparison remained architecture-wise and methodologically consistent.

8. Frozen 15-pipeline external comparison design

The user explicitly chose architecture-wise external comparison rather than testing only one global winner. Therefore the external study froze three pipelines per backbone: Phase 1 early stopping, Phase 2A standardized fixed-30, and Phase 2B tuned/changed configuration. The exact CNN-LSTM checkpoint, scaler, PCA model, SVM and internal-results artifact were hashed before external evaluation.

Backbone	Phase 1	Phase 2A	Phase 2B
VGG16	early stop, ep16	baseline best, ep30	lr_5e-4 best, ep27
VGG19	early stop, ep18	baseline best, ep30	dropout0.5 epoch30
ResNet50	early stop, ep7	baseline best, ep30	dropout0.2 best, ep28
ResNet152	early stop, ep8	baseline best, ep10	LSTM512 best, ep26
EfficientNet-B3	early stop, ep6	baseline best, ep27	LSTM512 best, ep29

External evaluation lock
After the frozen_15_manifest.json and frozen_15_selection.csv were created, the external dataset was not allowed to trigger checkpoint, hyperparameter or pipeline reselection. Poor external performance is treated as evidence, not as a reason to swap models.

Frozen artifact paths: `phase2/external_validation/frozen_15_manifest.json and frozen_15_selection.csv`

9. BDNeuro V8 selection, provenance review and overlap audit

9.1 Why BDNeuro was selected

The guide's practical question was: how well does the model perform on an MRI image that was not in the original dataset? A labeled external four-class MRI cohort is stronger evidence than arbitrary web images because it provides known class labels and enough cases for quantitative metrics. BDNeuro V8 matched the four project classes and contained 5,941 class-folder MRI images across its published train/val/test directories.

9.2 Why the full 5,941 images were initially eligible

The 15 project pipelines were already frozen and had never been trained, tuned or fit on BDNeuro. Therefore, from the project model's perspective, BDNeuro train/val/test were all unseen and could be merged for external evaluation. The original split labels were preserved only for traceability.

9.3 Provenance review vs empirical overlap evidence

The documented provenance chains did not reveal a clear Nickparvar-to-BDNeuro source bridge. Nickparvar is associated with Figshare/Cheng, SARTAJ and Br35H, while BDNeuro V8 was documented as a Bangladeshi clinical dataset. However, provenance metadata alone cannot establish image-level independence. The direct audit therefore took precedence.

9.4 Cross-dataset audit findings

Audit measure	Result
Internal images compared	7,023
BDNeuro class-folder MRIs	5,941
Cross-dataset exact file duplicate pairs	3,421
Cross-dataset exact decoded-pixel duplicate pairs	3,519
Unique BDNeuro exact-file-overlap images	3,295
Unique BDNeuro exact-pixel-overlap images	3,297
pHash <=6 candidate pairs	9,981
pHash <=4 strong candidate pairs	6,664
Within-BDNeuro exact file duplicates	0
Within-BDNeuro exact pixel duplicates	0

Why exact matches were removed despite different documented provenance
External validation is intended to measure performance on images not already present in the development corpus. Exact decoded-pixel identity is objective evidence that an image is not independent for this purpose, regardless of filename or claimed source. The historical mechanism of reuse/repackaging was not inferred because the available metadata did not establish it.

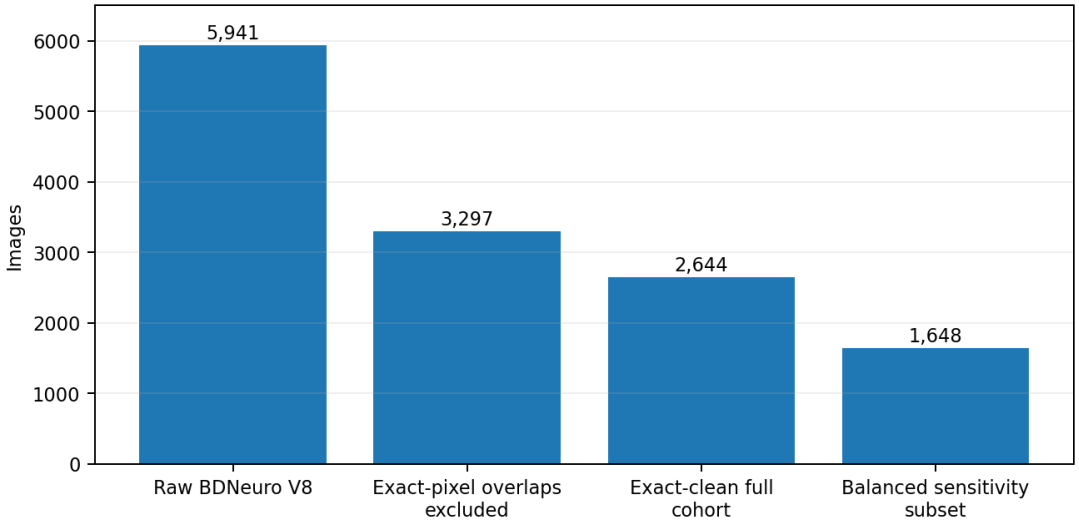


Figure 1. BDNeuro external-cohort construction and balanced sensitivity subset.

10. Exact-clean and balanced external cohorts

10.1 Locked exclusion rule

The final project decision was deliberately narrow: **remove exact decoded-pixel cross-dataset matches and retain all other BDNeuro images.** Perceptual-hash similarity was treated as a screening signal, not as automatic proof of duplication. This reflects the fact that axial brain MRIs naturally share anatomy, orientation, background and intensity structure.

Step	Images
Raw BDNeuro V8 MRI cohort	5,941
Unique exact decoded-pixel overlaps excluded	3,297
Exact-pixel-clean full external cohort	2,644

10.2 Residual perceptual-similarity finding

Among the 2,644 exact-pixel-clean images, 1,031 unique images had at least one pHash <=6 candidate against the internal corpus; 801 had at least one pHash <=4 candidate. These were retained because non-identical perceptual similarity alone was not accepted as evidence of duplicate identity. This is a pragmatic scope decision and residual near-duplicate risk should be acknowledged.

10.3 Full cohort class distribution

Class	Kept	Excluded exact overlaps	Kept share
Glioma	1,159	1,032	43.8%
Meningioma	472	1,006	17.9%
Notumor	412	375	15.6%
Pituitary	601	884	22.7%
Total	2,644	3,297	100%

10.4 Deterministic balanced sensitivity subset

Because the exact-clean full cohort is imbalanced, a secondary balanced sensitivity subset was created before prediction review. The smallest class contained 412 images, so the subset used exactly 412 per class = 1,648 images. Selection used deterministic SHA-256 ranking and did not use model outputs or performance values.

Class	Balanced images
Glioma	412
Meningioma	412
Notumor	412
Pituitary	412
Total	1,648

Reporting hierarchy
Primary external validation = full 2,644-image exact-overlap-clean cohort. Secondary sensitivity analysis = 1,648-image balanced subset. These are not two independent datasets because the balanced set is contained within the full cohort.

11. Completed external-validation protocol

Both external cohorts were evaluated through the same 15 pre-frozen pipelines. The external dataset was transform-only. There was no neural training, fine-tuning, checkpoint selection, scaler fitting, PCA fitting, SVM fitting, hyperparameter tuning or recalibration.

Stage	External inference action
Image preprocessing	Same deterministic evaluation preprocessing used by the project
CNN-LSTM	Load frozen checkpoint; inference only
Dense128 feature	Extract frozen 128-D representation
StandardScaler	saved_scaler.transform() only
PCA64	saved_pca.transform() only
Linear SVM	saved_svm.predict() / decision output
Metrics	Accuracy, macro precision/recall/F1, weighted F1, classification report, confusion matrix, per-image predictions

11.1 Why no external model selection was allowed

External validation answers whether internally selected models transfer to a new source domain. If BDNeuro performance were used to swap checkpoints or hyperparameters, BDNeuro would become another validation set and the external estimate would be biased.

11.2 Why both full and balanced cohorts were evaluated

The full cohort preserves all eligible external images and is the primary generalization test. The balanced subset asks whether conclusions change when every diagnostic class contributes equally. In a balanced four-class single-label problem, accuracy equals macro recall and weighted F1 equals macro F1; the observed balanced outputs satisfy these expected identities.

12. Completed external-validation results - full cohort

Primary cohort: n=2,644 exact-pixel-clean BDNeuro V8 images. The table below is the frozen 15-model result set.

Backbone	Stage	Ext. acc.	Macro P	Macro R	Macro F1	Internal->external acc.
VGG16	Phase 1	77.61%	77.66%	77.62%	75.50%	-17.32 pp
VGG16	Phase 2A	80.07%	80.87%	80.89%	78.71%	-15.82 pp
VGG16	Phase 2B	77.57%	78.16%	78.80%	76.01%	-18.93 pp
VGG19	Phase 1	75.76%	76.39%	78.04%	74.80%	-18.73 pp
VGG19	Phase 2A	79.27%	78.79%	80.86%	78.12%	-16.35 pp
VGG19	Phase 2B	75.00%	76.32%	77.20%	74.27%	-20.54 pp
ResNet50	Phase 1	81.09%	80.56%	83.92%	79.93%	-16.46 pp
ResNet50	Phase 2A	81.73%	82.50%	85.46%	81.42%	-16.96 pp
ResNet50	Phase 2B	86.04%	84.27%	87.96%	84.83%	-13.08 pp
ResNet152	Phase 1	81.28%	81.08%	85.18%	80.84%	-16.97 pp
ResNet152	Phase 2A	88.31%	86.43%	90.16%	87.85%	-10.29 pp
ResNet152	Phase 2B	86.54%	84.59%	88.98%	85.67%	-12.76 pp
EfficientNet-B3	Phase 1	83.09%	82.55%	85.40%	81.93%	-15.42 pp
EfficientNet-B3	Phase 2A	86.65%	85.12%	87.87%	85.12%	-12.65 pp
EfficientNet-B3	Phase 2B	86.57%	85.62%	88.76%	85.53%	-12.81 pp

Highest measured full-cohort result
ResNet152 Phase 2A: 88.31% external accuracy, 90.16% macro recall, 87.85% macro F1, with a -10.29 pp internal-to-external accuracy gap.

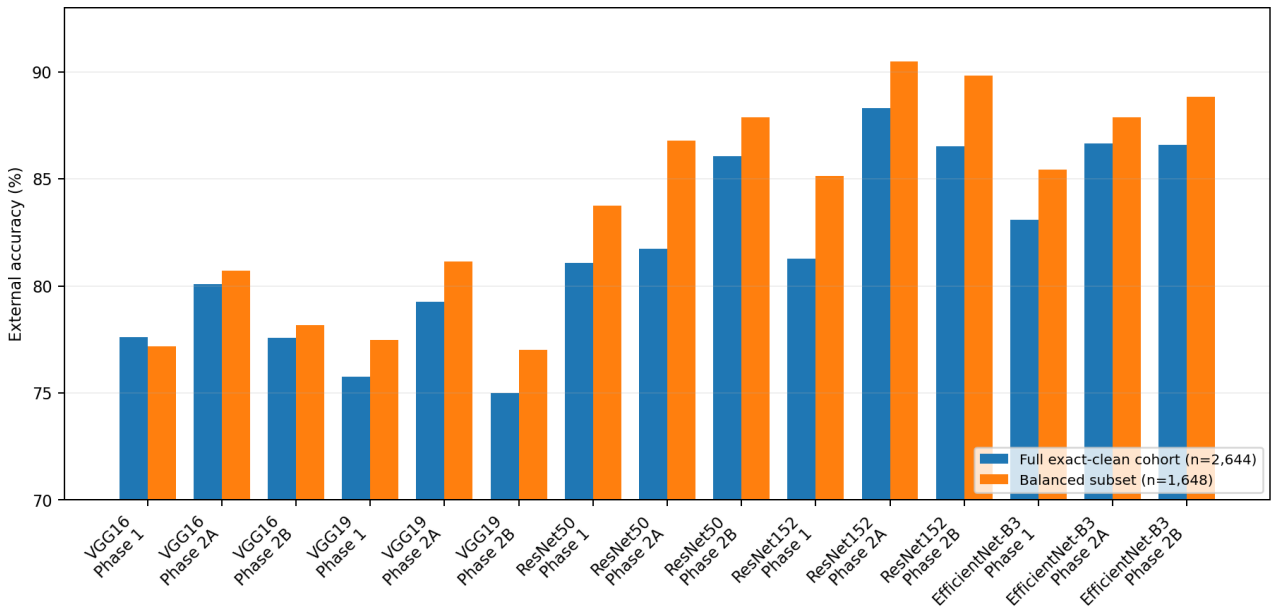


Figure 2. External accuracy for all 15 frozen pipelines on the full cohort and balanced subset.

13. Completed external-validation results - balanced subset

Secondary sensitivity cohort: n=1,648, exactly 412 images per class. It is a deterministic subset of the full cohort and therefore not an independent second external dataset.

Backbone	Stage	Ext. acc.	Macro P	Macro R	Macro F1	Internal->external acc.
VGG16	Phase 1	77.18%	79.41%	77.18%	76.82%	-17.74 pp
VGG16	Phase 2A	80.70%	83.31%	80.70%	80.62%	-15.18 pp
VGG16	Phase 2B	78.16%	80.90%	78.16%	77.88%	-18.35 pp
VGG19	Phase 1	77.49%	80.19%	77.49%	77.64%	-17.00 pp
VGG19	Phase 2A	81.13%	82.76%	81.13%	81.24%	-14.50 pp
VGG19	Phase 2B	77.00%	79.72%	77.00%	77.15%	-18.54 pp
ResNet50	Phase 1	83.74%	86.73%	83.74%	83.80%	-13.81 pp
ResNet50	Phase 2A	86.77%	89.12%	86.77%	86.90%	-11.92 pp
ResNet50	Phase 2B	87.86%	89.21%	87.86%	87.87%	-11.26 pp
ResNet152	Phase 1	85.13%	87.42%	85.13%	85.13%	-13.12 pp
ResNet152	Phase 2A	90.47%	90.85%	90.47%	90.50%	-8.13 pp
ResNet152	Phase 2B	89.81%	90.58%	89.81%	89.76%	-9.49 pp
EfficientNet-B3	Phase 1	85.44%	88.03%	85.44%	85.60%	-13.08 pp
EfficientNet-B3	Phase 2A	87.86%	89.19%	87.86%	87.79%	-11.44 pp
EfficientNet-B3	Phase 2B	88.83%	91.02%	88.83%	88.96%	-10.55 pp

Highest measured balanced result
ResNet152 Phase 2A: 90.47% accuracy and macro recall, 90.50% macro F1, with an -8.13 pp internal-to-external accuracy gap.

13.1 Mean performance across five backbones

Stage	Full accuracy	Full macro F1	Balanced accuracy	Balanced macro F1
Phase 1	79.77%	78.60%	81.80%	81.80%
Phase 2A	83.21%	82.25%	85.39%	85.41%
Phase 2B	82.34%	81.26%	84.33%	84.33%

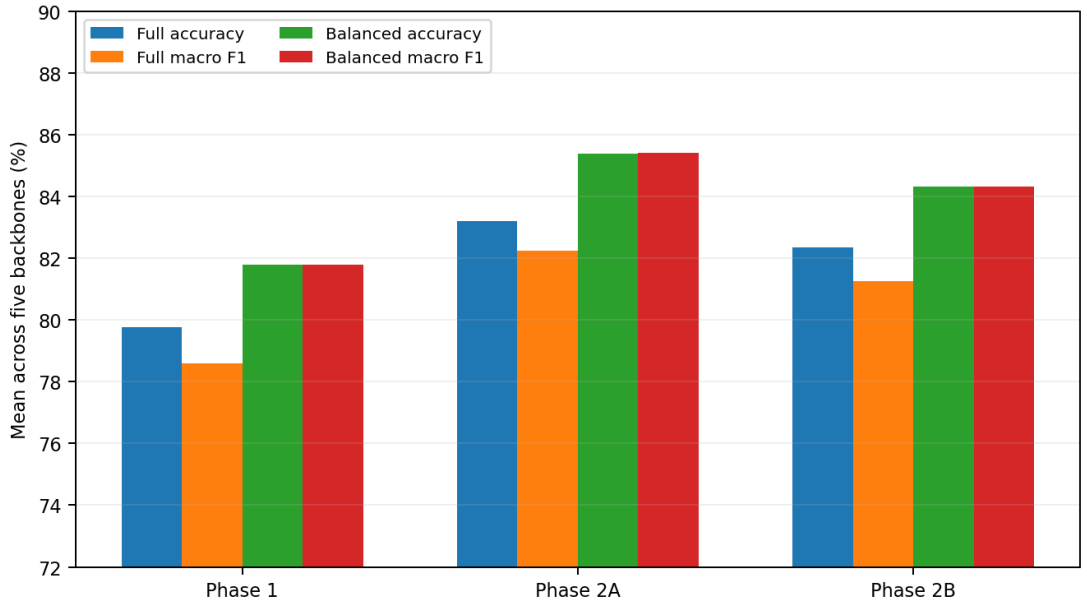


Figure 3. Mean external performance across five backbones by project stage.

14. Integrated external-validation analysis

14.1 Generalization gap

Every frozen pipeline performed worse externally than on the internal validation split. Full-cohort internal-to-external accuracy drops ranged from -10.29 pp to -20.54 pp. Balanced-subset drops ranged from -8.13 pp to -18.54 pp. This is direct evidence that the 95-99% internal validation range did not fully transfer under dataset/domain shift.

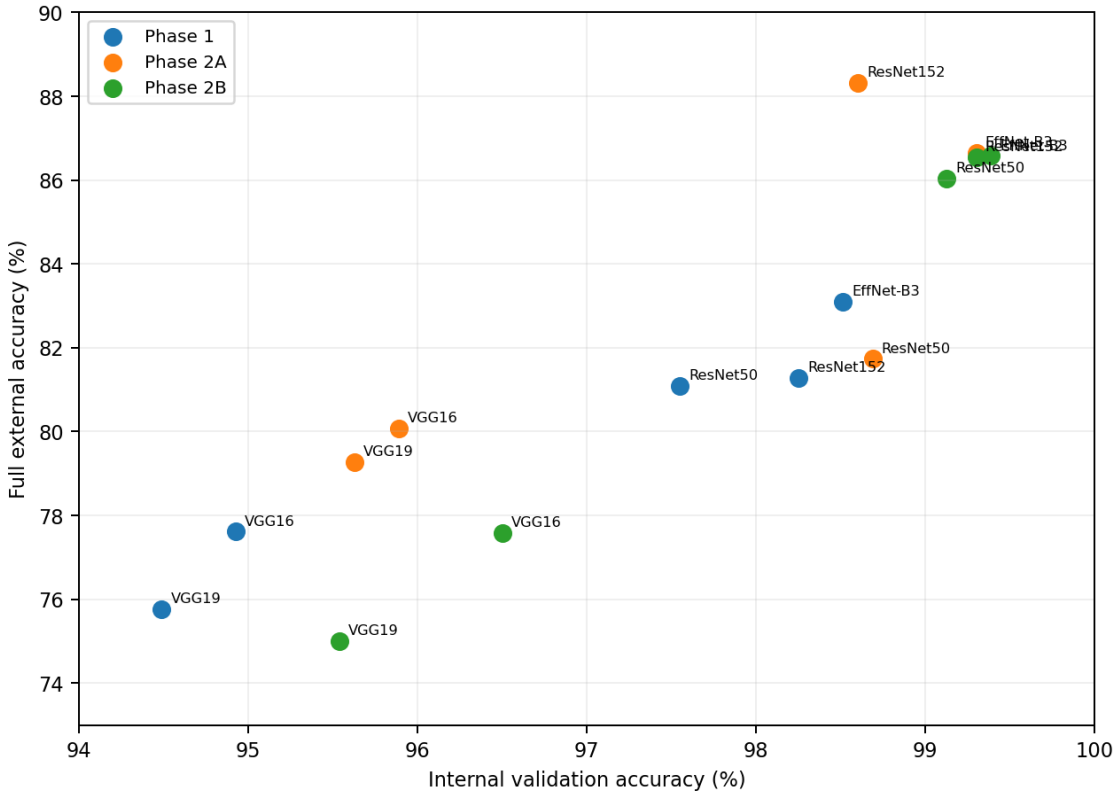


Figure 4. Internal hybrid validation accuracy versus full external accuracy. The dashed diagonal represents equal internal and external performance.

14.2 Phase 2A generalized better on average than Phase 2B

Across the five backbones, mean full-cohort accuracy was 79.77% for Phase 1, 83.21% for Phase 2A and 82.34% for Phase 2B. On the balanced subset the corresponding means were 81.80%, 85.39% and 84.33%. Therefore the standardized Phase 2A protocol generalized better on average than the internally tuned Phase 2B representatives.

Interpretation
This does not mean hyperparameter tuning was useless. It means internal tuning gains were architecture-dependent and did not consistently transfer to the external domain. External results were not used to revise the frozen choices.

15. Architecture-wise Phase 1 -> Phase 2A -> Phase 2B behavior

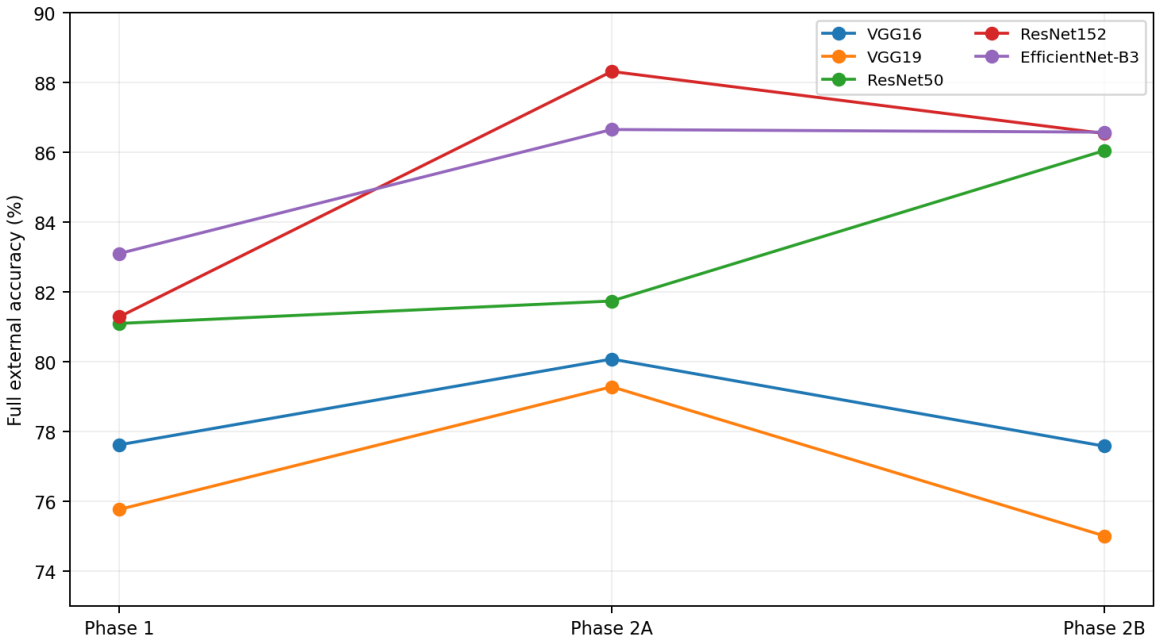


Figure 5. Full external accuracy trajectory across project stages for each backbone.

15.1 Full-cohort Phase 2B change relative to Phase 2A

Backbone	Phase 2A acc.	Phase 2B acc.	P2B - P2A
VGG16	80.07%	77.57%	-2.50 pp
VGG19	79.27%	75.00%	-4.27 pp
ResNet50	81.73%	86.04%	+4.31 pp
ResNet152	88.31%	86.54%	-1.78 pp
EfficientNet-B3	86.65%	86.57%	-0.08 pp

15.2 Balanced-subset Phase 2B change relative to Phase 2A

Backbone	Phase 2A acc.	Phase 2B acc.	P2B - P2A
VGG16	80.70%	78.16%	-2.55 pp
VGG19	81.13%	77.00%	-4.13 pp
ResNet50	86.77%	87.86%	+1.09 pp
ResNet152	90.47%	89.81%	-0.67 pp
EfficientNet-B3	87.86%	88.83%	+0.97 pp

VGG16 / VGG19
Phase 2A was stronger externally. Phase 2B changed configurations did not transfer their internal advantages and were notably weaker for VGG19.

ResNet50
The clearest Phase 2B external success. Dropout 0.2 improved full external accuracy from 81.73% to 86.04% and balanced accuracy from 86.77% to 87.86%.

ResNet152
Phase 2A was the strongest measured external configuration: 88.31% full and 90.47% balanced. LSTM512 Phase 2B was lower externally despite stronger internal validation.

EfficientNet-B3
Phase 2A and Phase 2B were very close on the full cohort; Phase 2B had slightly higher macro F1 and a clearer advantage on the balanced subset.

16. What the balanced sensitivity analysis adds

Most pipelines improved on the balanced subset relative to the full cohort, although VGG16 Phase 1 decreased slightly. The largest accuracy increase was ResNet50 Phase 2A: +5.04 pp. This indicates that the full cohort's class distribution and which images were retained after overlap cleaning influence aggregate accuracy.

Backbone	Stage	Balanced - full accuracy
VGG16	Phase 1	-0.43 pp
VGG16	Phase 2A	+0.64 pp
VGG16	Phase 2B	+0.58 pp
VGG19	Phase 1	+1.73 pp
VGG19	Phase 2A	+1.85 pp
VGG19	Phase 2B	+2.00 pp
ResNet50	Phase 1	+2.65 pp
ResNet50	Phase 2A	+5.04 pp
ResNet50	Phase 2B	+1.82 pp
ResNet152	Phase 1	+3.86 pp
ResNet152	Phase 2A	+2.16 pp
ResNet152	Phase 2B	+3.27 pp
EfficientNet-B3	Phase 1	+2.34 pp
EfficientNet-B3	Phase 2A	+1.22 pp
EfficientNet-B3	Phase 2B	+2.26 pp

Do not replace the primary result
The balanced subset is a sensitivity analysis, not a more correct external dataset. The full 2,644-image exact-clean cohort remains the primary external result because it preserves every eligible image after the locked exact-overlap exclusion rule.

17. Guide-ready interpretation and answers

Q1. "If I upload an MRI that was not in the dataset, how well does the model perform?"

The formal external experiment shows that performance drops substantially outside the original development corpus. Across 15 frozen pipelines, full-cohort accuracy ranged from 75.00% to 88.31%. The strongest measured full-cohort result was ResNet152 Phase 2A at 88.31% accuracy and 87.85% macro F1. On the balanced sensitivity subset it reached 90.47% accuracy and 90.50% macro F1. Therefore unseen-domain performance is meaningfully lower than the 95-99% internal validation range.

Q2. Why were 3,297 BDNeuro images removed even though the documented provenance looked different?

Because a direct image audit found exact decoded-pixel identity with images in the development corpus. External validation requires unseen images; exact pixel identity is objective evidence of overlap and takes precedence over provenance assumptions. The historical mechanism of reuse was not claimed because it was not established.

Q3. Why were pHash-similar images retained?

Perceptual similarity is not equivalent to exact identity. Brain MRIs naturally share anatomy, orientation, grayscale distribution and background. The locked rule therefore excluded exact decoded-pixel matches only. Residual near-duplicate risk is acknowledged rather than hidden.

Q4. Why test both full and balanced cohorts?

The full cohort measures performance on every eligible external image. The balanced subset checks whether conclusions remain similar when each class contributes equally. The balanced result is a sensitivity analysis of the same external cohort, not an independent second dataset.

Q5. Why not change models after seeing the external results?

Because doing so would convert the external dataset into another validation set. The 15 exact pipelines were frozen and artifact-hashed before external evaluation, so external success and failure remain unbiased evidence about transfer.

18. Scientific limitations and interpretation rules

Limitation	Required interpretation
Internal cross-split duplication	Historical Phase 1 / Phase 2 internal metrics may be optimistic.
Patient identity unavailable	Neither internal nor BDNeuro evaluation establishes patient-level independence.
Single seed	Phase 2A/2B differences have no repeated-seed variance estimate.
Phase 2 multiple comparisons	Phase 2B representatives were chosen after many internal configurations/states; selection optimism is possible.
Only two retained states per Phase 2B run	Best-neural vs epoch30 analysis does not identify the absolute best downstream epoch among all 30.
BDNeuro exact-only exclusion rule	Non-identical pHash candidates were retained; residual near-duplicate risk remains.
Balanced set is nested in full set	Do not present full and balanced results as two independent external validations.
External site/sequence differences	Observed drops reflect a mixture of domain shift, class mix, preprocessing/acquisition differences and dataset characteristics; causation cannot be assigned to one factor.
External per-class synthesis not re-audited here	This document analyzes the verified master CSV aggregate metrics; detailed per-class external confusion artifacts should be reviewed separately before final thesis claims.
Prototype status	Research/educational only; not clinical validation or a medical device.
Explainability qualitative	Grad-CAM and occlusion are not tumor segmentation or clinical reasoning proof.

Terminology that must remain consistent

- Use "neural-head vs PCA+SVM downstream-classifier comparison"; do not call it a PCA-only ablation.

- Call the new external set "BDNeuro V8 exact-overlap-clean external cohort" or equivalent; do not claim complete independence.

- Call the 1,648-image set a "balanced sensitivity subset"; it is nested within the 2,644-image full cohort.

- Describe original standardized metrics as historical contaminated-split results after the leakage audit.

- Do not claim Phase 2B tuning universally improves external performance.

- Do not claim statistical significance from single-seed differences.

- Do not use external results to retroactively select or retrain the frozen 15 pipelines.

19. Current project state and future work

Status as of 3 Oct 2026
The project has completed its formal architecture-wise external validation at the aggregate-metric level. The next optional scientific work is per-class external error synthesis, the planned 40-image web stress test, and an explicit decision about duplicate-group-aware internal re-splitting/retraining.

19.1 Planned in-the-wild web-image stress test - not yet performed

Purpose: directly demonstrate the guide's "random unseen MRI from the web" scenario while keeping it separate from formal external validation.

Design item	Locked plan
Total	40 source-verified images
Per class	10 glioma, 10 meningioma, 10 pituitary, 10 no-tumor
Discovery	Google Images may be used to find candidates, but the original medical/case page must establish diagnosis
Sources	Multiple independent radiology teaching cases, hospitals/universities or open-access case reports
Record	URL, source diagnosis, sequence if known, date accessed, crop/annotation status, image file, all 15 predictions
Audit	Exact overlap check against original 7,023 images before accepting the final 40
Interpretation	Exploratory / in-the-wild stress test; descriptive X/40 and macro metrics only, not a clinical real-world accuracy claim

19.2 Duplicate-group-aware re-splitting/retraining - unresolved stronger remediation

The scientifically stronger response to the original internal leakage audit is to construct exact/near-duplicate groups, split groups rather than individual images, retrain all five backbones from scratch, and refit each scaler/PCA/SVM using only the new training split. This remains unresolved and would be needed before treating internal test performance as a clean independent-generalization estimate.

20. ChatGPT / Claude transfer instructions

20.1 Authority rule

Treat this document as the project-status snapshot as of 3 October 2026. If an older report, PPT or conversation conflicts with it, prefer the newer verified external master CSVs, current overlap-cleaning decisions and current frozen-pipeline status recorded here.

20.2 What future ChatGPT / Claude work must preserve

- Five-backbone final scope: VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3.

- Architecture: MRI -> CNN -> spatial sequence -> LSTM -> BN -> Dropout -> Dense128/ReLU -> neural head; Dense128 features -> StandardScaler -> PCA64 -> linear SVM.

- Baseline RMSprop, LR1e-4, batch8, LSTM256, dropout0.3, training-only augmentation, deterministic evaluation; VGG16/19 frozen.

- Internal leakage audit is complete and materially changes interpretation of historical metrics.

- Phase 2A fixed-30 training and 10/10 downstream retained-state evaluations are complete.

- Phase 2B 35/35 DL runs and 70/70 full-hybrid retained-state evaluations are complete.

- Fifteen exact external pipelines were frozen before external evaluation; do not reselect using BDNeuro.

- BDNeuro V8 raw 5,941 -> exclude 3,297 unique exact decoded-pixel overlaps -> full exact-clean n=2,644.

- Balanced sensitivity subset n=1,648, 412/class, deterministic and nested within full cohort.

- External validation is complete for both full and balanced cohorts. Phase 2A generalized better on average than Phase 2B; ResNet50 is the clearest Phase 2B external improvement.

- 40-image web stress test remains pending.

- Claude is preferred for PPT/document generation/editing; ChatGPT handles code, experiments, methodology, auditing and scientific interpretation.

20.3 Mistakes to avoid

- Saying external validation is still pending.

- Calling the 5,941-image raw BDNeuro release the evaluated external cohort.

- Claiming no overlap simply because the documented dataset provenance differs.

- Claiming the 2,644-image exact-clean cohort is proven near-duplicate-free or patient-independent.

- Calling the balanced subset an independent second external dataset.

- Choosing a new model because it looked better on BDNeuro.

- Claiming Phase 2B tuning universally improved generalization.

- Calling Grad-CAM/occlusion segmentation or clinical trust evidence.

21. Compact handoff block

PROJECT: DL TUMOR CLASSIFICATION. STATUS: 03 Oct 2026.

Five backbones: VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3. Architecture: MRI -> CNN spatial grid -> spatial sequence -> LSTM -> BN -> Dropout -> Dense128/ReLU -> neural head; 128-D features -> StandardScaler -> PCA64 -> linear SVM. Baseline RMSprop, LR1e-4, batch8, LSTM256, dropout0.3, seed42; train augmentation only; eval deterministic; VGG16/19 frozen.

Phase 1 complete. Neural test acc: 93.67, 93.14, 97.33, 98.17, 98.02%. PCA+SVM test acc: 94.97, 93.52, 98.09, 98.40, 98.47%. Internal duplicate audit complete: 7,023 images; 464 exact cross-split pairs + 98 high-confidence near pairs = 562 material pairs; 250 official-test images implicated; material_clean n=1,061; patient independence unknown.

Phase 2A complete: 5 fixed 30-epoch baselines + 10/10 full-hybrid retained-state evaluations. Frozen Phase2A representatives by hybrid validation: VGG16 ep30 95.888%, VGG19 ep30 95.626%, ResNet50 ep30 98.688%, ResNet152 ep10 98.600%, EffNet-B3 ep27 99.300%.

Phase 2B complete: 35/35 DL runs + 70/70 retained-state full-hybrid evaluations. Frozen changed-config reps: VGG16 lr5e-4 best ep27 96.500%; VGG19 dropout0.5 epoch30 95.538%; ResNet50 dropout0.2 best ep28 99.125%; ResNet152 LSTM512 best ep26 99.300%; EffNet-B3 LSTM512 best ep29 99.388%. Across 35 configs, best-neural retained state had higher downstream val in 20, epoch30 in 8, identical in 7.

Pre-external freeze complete: 15 exact pipelines (Phase1/2A/2B x five backbones), artifact hashes recorded. BDNeuro V8 chosen for formal external validation. Raw class-folder MRIs=5,941. Cross-dataset audit vs original 7,023 found 3,297 unique exact decoded-pixel overlaps; exact matches excluded. pHash similarity used as screening only and not automatic exclusion. Full exact-clean cohort n=2,644: glioma1159, meningioma472, notumor412, pituitary601. Balanced deterministic sensitivity subset n=1,648: 412/class.

External validation COMPLETE for all 15 pipelines on both cohorts, with no external fitting/tuning/reselection. Full external accuracy range 75.00-88.31%; balanced 77.00-90.47%. Highest measured full and balanced accuracy: ResNet152 Phase2A, 88.31% full with macroF1 87.85%, 90.47% balanced with macroF1 90.50%. Mean external accuracy by stage: full Phase1 79.77, Phase2A 83.21, Phase2B 82.34%; balanced 81.80, 85.39, 84.33%. Every pipeline dropped from internal validation externally. Phase2B transferred inconsistently: ResNet50 improved clearly; VGG16/VGG19/ResNet152 were worse than Phase2A externally; EffNet-B3 was close on full and Phase2B better on balanced.

Formal external validation should be reported as full exact-clean primary cohort plus balanced sensitivity subset, NOT two independent datasets. Web stress test remains pending: 40 source-verified web/case MRI images, 10/class. Duplicate-group-aware internal re-split/retrain remains unresolved. GUI/Grad-CAM/SVM occlusion complete; prototype only; explainability qualitative.

Appendix A. Full 15-model external result tables

A1. Full exact-clean cohort (n=2,644)

Stage	Backbone	Variant	State	Ep	Int val acc	Ext acc	Macro F1
Phase 1	VGG16	early_stopping	best_model	16	94.926	77.610	75.499
Phase 2A	VGG16	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	95.888	80.068	78.710
Phase 2B	VGG16	lr_5e-4	best_val_within_30	27	96.500	77.572	76.011
Phase 1	VGG19	early_stopping	best_model	18	94.488	75.756	74.798
Phase 2A	VGG19	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	95.626	79.274	78.123
Phase 2B	VGG19	dropout_0.5	epoch_30	30	95.538	75.000	74.268
Phase 1	ResNet50	early_stopping	best_model	7	97.550	81.089	79.935
Phase 2A	ResNet50	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	98.688	81.732	81.421
Phase 2B	ResNet50	dropout_0.2	best_val_within_30	28	99.125	86.044	84.834
Phase 1	ResNet152	early_stopping	best_model	8	98.250	81.278	80.842
Phase 2A	ResNet152	lr1e-4_lstm256_dropout0.3	best_val_within_30	10	98.600	88.313	87.854
Phase 2B	ResNet152	lstm_512	best_val_within_30	26	99.300	86.536	85.672
Phase 1	EfficientNet-B3	early_stopping	best_model	6	98.513	83.094	81.933
Phase 2A	EfficientNet-B3	lr1e-4_lstm256_dropout0.3	best_val_within_30	27	99.300	86.649	85.123
Phase 2B	EfficientNet-B3	lstm_512	best_val_within_30	29	99.388	86.573	85.526

A2. Balanced sensitivity subset (n=1,648)

Stage	Backbone	Variant	State	Ep	Int val acc	Ext acc	Macro F1
Phase 1	VGG16	early_stopping	best_model	16	94.926	77.184	76.818
Phase 2A	VGG16	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	95.888	80.704	80.619
Phase 2B	VGG16	lr_5e-4	best_val_within_30	27	96.500	78.155	77.885
Phase 1	VGG19	early_stopping	best_model	18	94.488	77.488	77.640
Phase 2A	VGG19	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	95.626	81.129	81.237
Phase 2B	VGG19	dropout_0.5	epoch_30	30	95.538	77.002	77.154
Phase 1	ResNet50	early_stopping	best_model	7	97.550	83.738	83.803
Phase 2A	ResNet50	lr1e-4_lstm256_dropout0.3	best_val_within_30	30	98.688	86.772	86.898
Phase 2B	ResNet50	dropout_0.2	best_val_within_30	28	99.125	87.864	87.873
Phase 1	ResNet152	early_stopping	best_model	8	98.250	85.133	85.132
Phase 2A	ResNet152	lr1e-4_lstm256_dropout0.3	best_val_within_30	10	98.600	90.473	90.497
Phase 2B	ResNet152	lstm_512	best_val_within_30	26	99.300	89.806	89.762
Phase 1	EfficientNet-B3	early_stopping	best_model	6	98.513	85.437	85.605
Phase 2A	EfficientNet-B3	lr1e-4_lstm256_dropout0.3	best_val_within_30	27	99.300	87.864	87.787
Phase 2B	EfficientNet-B3	lstm_512	best_val_within_30	29	99.388	88.835	88.956

Appendix B. Phase 2 training sensitivity reference

These values preserve the 40-run fixed-30-epoch training-level sensitivity record from the prior report. They are neural validation trajectories, not external results.

VGG16

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp
baseline	1e-4	256	0.3	30	95.451	95.451	+0.000
lr_1e-5	1e-5	256	0.3	30	90.201	90.201	-5.249
lr_5e-5	5e-5	256	0.3	30	95.276	95.276	-0.175
lr_5e-4	5e-4	256	0.3	27	95.976	95.976	+0.525
lstm_128	1e-4	128	0.3	30	95.801	95.801	+0.350
lstm_512	1e-4	512	0.3	19	95.888	95.626	+0.437
dropout_0.2	1e-4	256	0.2	27	95.626	95.538	+0.175
dropout_0.5	1e-4	256	0.5	30	95.538	95.538	+0.087

VGG19

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp
baseline	1e-4	256	0.3	30	95.188	95.188	+0.000
lr_1e-5	1e-5	256	0.3	30	89.939	89.939	-5.249
lr_5e-5	5e-5	256	0.3	28	94.663	94.051	-0.525
lr_5e-4	5e-4	256	0.3	27	95.538	95.013	+0.350
lstm_128	1e-4	128	0.3	28	94.663	94.313	-0.525
lstm_512	1e-4	512	0.3	30	95.888	95.888	+0.700
dropout_0.2	1e-4	256	0.2	29	94.838	94.751	-0.350
dropout_0.5	1e-4	256	0.5	28	95.363	95.188	+0.175

ResNet50

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp
baseline	1e-4	256	0.3	30	98.425	98.425	+0.000
lr_1e-5	1e-5	256	0.3	27	96.938	96.763	-1.487
lr_5e-5	5e-5	256	0.3	23	98.600	98.425	+0.175
lr_5e-4	5e-4	256	0.3	25	94.751	90.989	-3.675
lstm_128	1e-4	128	0.3	28	98.425	96.675	+0.000
lstm_512	1e-4	512	0.3	29	99.038	98.250	+0.612
dropout_0.2	1e-4	256	0.2	28	99.125	98.513	+0.700
dropout_0.5	1e-4	256	0.5	26	98.338	97.463	-0.087

ResNet152

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp
baseline	1e-4	256	0.3	10	98.775	98.688	+0.000
lr_1e-5	1e-5	256	0.3	29	97.025	96.938	-1.750
lr_5e-5	5e-5	256	0.3	17	98.950	98.513	+0.175
lr_5e-4	5e-4	256	0.3	20	93.963	93.613	-4.812
lstm_128	1e-4	128	0.3	29	98.950	98.688	+0.175
lstm_512	1e-4	512	0.3	26	99.213	98.775	+0.437
dropout_0.2	1e-4	256	0.2	27	98.600	98.075	-0.175
dropout_0.5	1e-4	256	0.5	29	98.863	98.425	+0.087

EfficientNet-B3

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp
baseline	1e-4	256	0.3	27	99.300	98.688	+0.000
lr_1e-5	1e-5	256	0.3	28	96.413	96.325	-2.887
lr_5e-5	5e-5	256	0.3	14	98.950	98.950	-0.350
lr_5e-4	5e-4	256	0.3	27	98.425	97.638	-0.875
lstm_128	1e-4	128	0.3	30	99.213	99.213	-0.087
lstm_512	1e-4	512	0.3	29	99.388	98.425	+0.087
dropout_0.2	1e-4	256	0.2	15	98.950	98.688	-0.350
dropout_0.5	1e-4	256	0.5	26	99.125	98.688	-0.175

Appendix C. Evidence, scripts and artifact register

Artifact / path	What it supports
phase2/external_validation/frozen_15_manifest.json	Pre-external freeze; artifact hashes and exact 15-pipeline identities
phase2/external_validation/frozen_15_selection.csv	Human-readable frozen selection table
actual_training_scripts/evaluate_hyperparam_pca_svm.py	One Phase 2B retained-state full-hybrid validation evaluation
phase2/run_all_hyperparam_pca_svm.py	70-evaluation Phase 2B downstream queue
actual_training_scripts/compare_external_15.py	Frozen 15-pipeline external evaluator; named result roots for full vs balanced
actual_training_scripts/audit_bdneuro_overlap.py	Cross-dataset exact/pixel/pHash audit
actual_training_scripts/build_bdneuro_exact_clean_cohort.py	Creates 2,644-image derived exact-clean cohort; raw dataset preserved
actual_training_scripts/build_bdneuro_balanced_subset.py	Creates deterministic 1,648-image balanced subset
external_data/BDNeuro_V8_exact_clean	Primary external cohort
external_data/BDNeuro_V8_exact_clean_balanced	Secondary balanced sensitivity subset
phase2/external_validation/bdneuro_overlap_audit/	Cross-dataset audit CSV/JSON outputs
phase2/external_validation/bdneuro_exact_clean_cohort/	Kept/excluded manifests and cohort summary
phase2/external_validation/bdneuro_balanced_subset/	Balanced selection manifest and summary
phase2/external_validation/comparison_15_models_bdneuro_full/	Primary 15-model external master JSON/CSV and per-model outputs
phase2/external_validation/comparison_15_models_bdneuro_balanced/	Balanced 15-model sensitivity master JSON/CSV and per-model outputs
model_cnn_lstm.py / train_backbone*.py / evaluate_backbone*.py	Core architecture and standardized training/evaluation families
app.py + Grad-CAM / occlusion apps	Research GUI and qualitative explainability work

End-state summary
The project is now beyond Phase 2C at the aggregate external-validation level. The formal 15-pipeline external comparison is complete on both the 2,644-image exact-clean cohort and its 1,648-image balanced sensitivity subset. The next optional experiments are detailed external per-class error synthesis, the 40-image web stress test, and a decision on duplicate-group-aware internal retraining.